reabsorption from this point is dependent on the body's need and is regulated by hormones. Reabsorption of bicarbonate (HCO_3^-) takes place to regulate the blood pH. Homeostasis of K^+ and Na^+ in the blood is also regulated in this region.

Aquaporins are water–permeable channels (membrane transport proteins) that allow water to move across the epithelial cells in relation to the osmotic difference from the lumen to the interstitial fluid.

Collecting duct is permeable to water, secretes K^+ (potassium ions are actively transported into the tubule) and reabsorbs Na^+ to produce concentrated urine. The change in permeability to water is due to the presence of number of water-permeable channels called **aquaporins**.

Tubular secretion- Substances such as H^+, K^+, NH_4^+, creatinine and organic acids move into the filtrate from the peritubular capillaries into the tubular fluid. Most of the water is absorbed in the proximal convoluted tubule and Na^+ is exchanged for water in the loop of Henle. Hypotonic fluid enters the distal convoluted tubule and substances such as urea and salts pass from peritubular blood into the cells of DCT. The urine excreted contains both filtered and secreted substances. Once it enters the collecting duct, water is absorbed and concentrated hypertonic urine is formed. For every H^+ secreted into the tubular filtrate, a Na^+ is absorbed by the tubular cell. The H^+ secreted combines with HCO_3^-, HPO_3^- and NH_3^- and gets fixed as $H_2CO_4^+$, $H_2PO_4^+$ and NH_4^+ respectively. Since H^+ gets fixed in the fluid, reabsorption of H^+ is prevented.

Osmolarity - The solute concentration of a solution of water is known as the solutions osmolarity, expressed as milliosmoles /liter (mOsm/L)

Formation of concentrated urine
Formation of concentrated urine is accomplished by kidneys using counter current mechanisms. The major function of Henle's loop is to concentrate Na^+ and Cl^-. There is low osmolarity near the cortex and high osmolarity towards the medulla. This osmolarity in the medulla is due to the presence of the solute transporters and is maintained by the arrangement of the loop of Henle, collecting duct and vasa recta. This arrangement allows movement of solutes from the filtrate to the interstitial fluid. At the transition between the proximal convoluted tubule and the descending loop of Henle the osmolarity of the interstitial fluid is similar to that of the blood – about 300mOsm.

Ascending and descending limbs of Henle, create a counter current multiplier (interaction between flow of filtrate through the limbs of Henle's and JMN) by active transport. Figure 8.8 (a) shows the counter current multiplier created by the long loops of **Henle of the JM nephrons which creates medullary osmotic gradient.**

As the fluid enters the descending limb, water moves from the lumen into the interstitial fluid and the osmolarity of interstitial fluid decreases. To counteract this dilution the region of the ascending limb actively pumps solutes from the lumen into the interstitial fluid and the osmolarity increases to about 1200mOsm in medulla. This mismatch between water

XI Std Zoology Chapter-8 EM.indd 143 30/12/2022 13:48:07